

Note 12.5 Lines and Planes in Space

Lines and Line Segments in Space

Vector Equation for a Line

A vector equation for the line L through $P_0(x_0, y_0, z_0)$ parallel to $\mathbf{v}$ is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, -\infty < t < \infty$$

where $\mathbf{r}$ is the position vector of a point $P(x, y, z)$ on L and $\mathbf{r}_0$ is the position vector of $P_0(x_0, y_0, z_0)$

Parametric Equations for a Line

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is

$$x = x_0 + tv_1, y = y_0 + tv_2, z = z_0 + tv_3, -\infty < t < \infty$$

The Distance from a Point to a Line in Space

Distance from a Point S to a Line Through P Parallel to $\mathbf{v}$

$$d = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

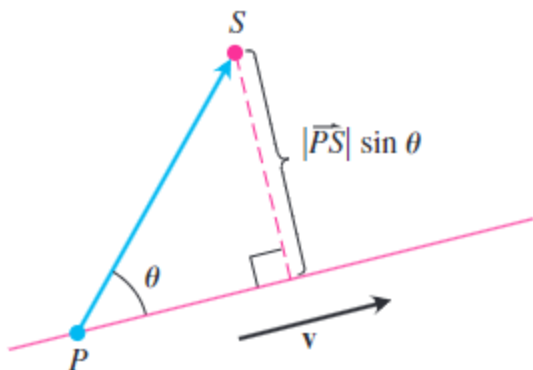


FIGURE 12.38 The distance from S to the line through P parallel to $\mathbf{v}$ is $|\overrightarrow{PS}| \sin \theta$, where θ is the angle between $\overrightarrow{PS}$ and $\mathbf{v}$.

An Equation for a Plane in Space

Equation for a Plane

The plane through $P_0(x_0, y_0, z_0)$ normal to $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ has

Vector equation:

$$\mathbf{n} \cdot \overrightarrow{P_0P} = 0$$

Component equation:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Simplified:

$$Ax + By + Cz = D, D = Ax_0 + By_0 + Cz_0$$

Lines of Intersection

Two planes are parallel if and only if their normals are parallel, or $\mathbf{n}_1 = k\mathbf{n}_2$ for some scalar k .

If two planes are intersected, then the formula can be obtained by associating the formula of two planes and its direction is perpendicular to the normals of both two planes.

The Distance from a Point to a Plane

If P is a point on the plane with normal $\mathbf{n}$, then the distance from any point S to the plane is the length of the vector projection of $\overrightarrow{PS}$ onto $\mathbf{n}$. That is,

$$d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$$

where $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is the normal to the plane.

Angles Between Planes

The angle between two intersecting planes is defined to be the acute angle between their normal vectors. That is,

$$\theta = \cos^{-1} \left(\frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|} \right)$$

Exercises 12.4

P712 9, 23, 35, 39, 59, 67